8.2 Bounding Without Solving

WORKSHEET

The question

STEP III 2013, Question 7

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/13-s3-q7

Hints

Try the question on your own first. If you get stuck, read the hints *one at a time*, and go back to the question after each one. Hints marked *Pause and try* are the key steps. There is space after each hint for your working.

Ask yourself For part (i), think of y as the position of a particle and x as time. Then $y'' = -y^3$ is Newton's law with force $-y^3$. What are the kinetic and potential energies?

STAGE 1 Conservation

Covers: Part (i)

Ask yourself Differentiate E with respect to x by the chain rule, and factor out $2y'$. What does the differential equation say about the bracket that is left?

Ask yourself So E is constant. Find its value at $x = 0$. Since $(y')^2 \geq 0$, what does that tell you about y^4?

STAGE 2 Dissipation

Covers: Part (ii)

Pause and try Differentiate E again, and use the equation to replace $v'' + \sinh v$. What sign does $\frac{\mathrm{dE}}{\mathrm{d}x}$ have for $x \geq 0$? Compare $\mathrm{E}(x)$ with $\mathrm{E}(0)$, and work out $\cosh(\ln 3)$.

STAGE 3 Inventing the energy

Covers: Part (iii)

Pause and try Try $\mathrm{E} = (w')^2 + F(w)$. Differentiate, and choose $F'(w)$ so that the equation can replace w'' and leave only a multiple of $(w')^2$. You will need a function whose derivative is $w\cosh w$: try differentiating $w\sinh w$.

Ask yourself To show $5\cosh x - 4\sinh x - 3 \geq 0$, write it using e^x and e^{-x} and look for a perfect square. Then compare $\mathrm{E}(x)$ with $\mathrm{E}(0)$. What sign does $w\sinh w$ have?

What's really going on

Lyapunov functions

In his 1892 doctoral thesis, Aleksandr Lyapunov studied the *stability* of motion: if a system is disturbed slightly from equilibrium, does it return, or wander away? His "second method" was to find a function, now called a *Lyapunov function*, that is smallest at the equilibrium and never increases along solutions. Then solutions that start near the equilibrium stay near it. The beauty of the method is that it needs no formula for the solutions, only the differential equation itself, which is exactly what this question demonstrates. Lyapunov functions are now used to prove the stability of everything from aircraft control systems to power grids to ecosystems.

Energy in physics and engineering

Part (i) is a nonlinear oscillator: like a spring, but with a restoring force proportional to the cube of the displacement. Unlike a simple harmonic oscillator, its period depends on its amplitude (larger oscillations are faster), but energy conservation still confines it. Part (ii) adds damping that grows with time, and energy drains away. In engineering, energy methods are used to show that structures will not shake themselves apart and that control systems will not oscillate out of control. In the theory of partial differential equations, "energy estimates" of this kind are the main tool for proving that solutions exist and behave well; questions about the equations of fluid flow, including one of the Millennium Prize Problems (the Navier–Stokes problem), come down to whether such bounds can be found.

Try this next

1. **Simple harmonic motion.** For $y'' + y = 0$ with $y(0) = A$, $y'(0) = 0$, find the conserved energy and deduce $|y| \le A$. Check against the exact solution.

2. **The pendulum.** The angle θ of a pendulum satisfies $\theta'' + \sin\theta = 0$ (in suitable units). Show that $(\theta')^2 - 2\cos\theta$ is constant. If the pendulum starts at rest at angle θ_0, where $|\theta_0| \le \pi$, show that $|\theta|$ never exceeds $|\theta_0|$. What goes wrong if $|\theta_0| > \pi$?

3. **Damping.** For $y'' + 2ky' + y = 0$ with $k > 0$, show that $(y')^2 + y^2$ decreases. Does it necessarily tend to zero?

4. **An escape.** For $y'' - y^3 = 0$ with $y(0) = 1$, $y'(0) = 0$, find a conserved quantity and show that y is not bounded by it. What goes wrong?

Further reading

- **Steven Strogatz, *Nonlinear Dynamics and Chaos*** (CRC Press). Chapters 6 and 7 cover phase planes, conservative systems and Lyapunov functions.
- **D. W. Jordan & P. Smith, *Nonlinear Ordinary Differential Equations*** (Oxford University Press). A standard undergraduate text with many examples of energy methods.
- **Keith Devlin, *The Millennium Problems*** (Basic Books). Includes an accessible account of the Navier–Stokes problem.

The full guided solution is in Section 8.2:
https://beyondstep.keithmcnulty.org/ch08/bounding-without-solving.html